

## Human Review: Finite-codimensional quasicomplement counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct as a literal counterexample.

## Human Comments

The proposed solution appears correct as an answer to Problem 2.1 exactly as stated in Barbosa–Raposo–Ribeiro’s paper. The example is

$$X = \ell_2 \oplus_2 \mathbb{R}, \quad Y = \ell_2 \oplus \{0\}, \quad F = \{0\} \oplus \mathbb{R}.$$

Then  $Y$  is closed, infinite-dimensional, and quasicomplemented. In fact, it is 1-complemented, so the example is very elementary.

The obstruction is finite codimension. Since

$$X = Y \oplus F,$$

any subspace  $Z$  which is disjoint from  $Y$  and contains  $F$  must equal  $F$ . Indeed, if  $z \in Z$ , write  $z = y + f$  with  $y \in Y$  and  $f \in F$ . Since  $F \subset Z$ , one has  $y = z - f \in Z \cap Y = \{0\}$ , so  $z = f \in F$ . Hence  $Z = F$ , and therefore

$$\dim Z/F = 0,$$

not infinity.

## Interpretation

Mathematically, the argument is correct. However, this seems likely to be a minor issue in the formulation of the question rather than a substantial counterexample to the intended problem. The example is simply a finite-codimensional complemented subspace, and the obstruction comes from the fact that the chosen finite-dimensional  $F$  already exhausts the quotient  $X/Y$ . There is then no room left for a quasicomplement  $Z$  containing  $F$  with infinite-dimensional quotient  $Z/F$ .

This strongly suggests that the intended question was probably meant in the setting of infinite codimension, which is also the setting emphasized by the paper’s later positive results.

## Literature Check

A preliminary online check did not find a later answer to this exact point. OpenAlex reports no citing works for arXiv:2403.05806, a direct OpenAlex citing-works query returned no results, and OpenCitations returned an empty citation list for the arXiv DOI. Targeted searches for the terminology around total  $\alpha$ -quasicomplementation, finite codimension, and Problem 2.1 did not reveal another source discussing this example.

## Review Outcome

The packet should be recorded as a correct literal counterexample, but with the caveat that it probably identifies a missing hypothesis or intended restriction in the question. I would not present it as a deep mathematical contribution; rather, it is a semantic warning about finite-codimensional cases.